CH1101: Quantum Chemistry & Spectroscopy (24) 28/09/2026

Lec13

Particle in a sphere

$$\psi_{n,l,m_l}(r,\theta,\phi) = R_{n,l}(r)\, Y_{l,m_l}(\theta,\phi)$$

Polar coordinate

$R_{n,l}(r)$: Radial part

$Y_{l,m_l}(\theta,\phi)$: Angular part

ψ_{n,l,m_l}: Wavefunction "Atomic Orbitals"

FOR

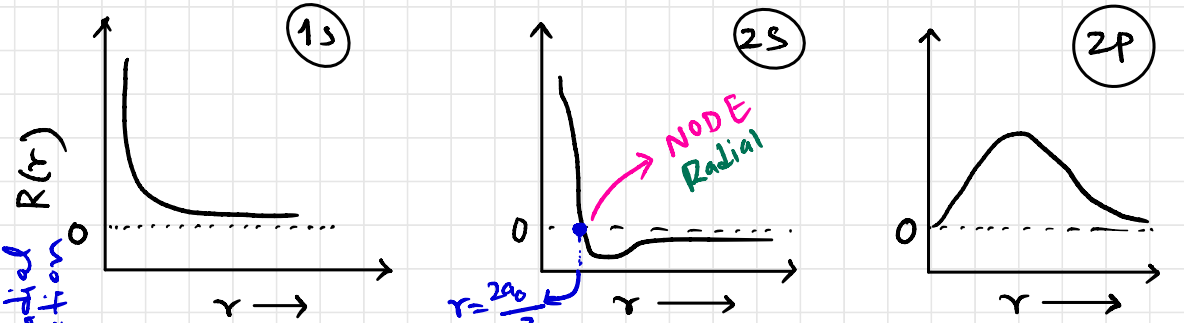
$0 \leq r < \infty$

$0 \leq \theta \leq \pi$

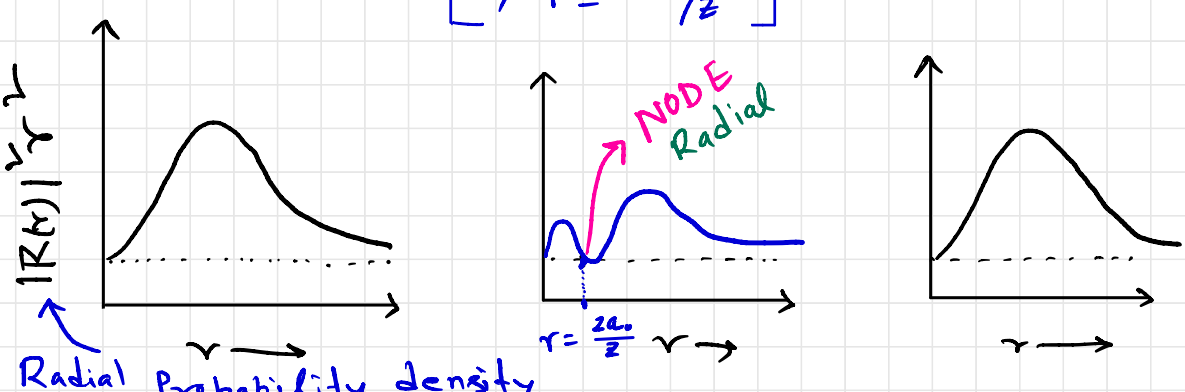
$0 \leq \phi \leq 2\pi$

Wavefunction	Expression
ψ_{100}	$\psi_{1s} = C\, e^{-Zr/a_0}$
ψ_{200}	$\psi_{2s} = C\left(2 - \frac{Zr}{a_0}\right) e^{-Zr/2a_0}$
ψ_{210}	$\psi_{2p_z} = C\left(\frac{Zr}{a_0}\right) e^{-Zr/2a_0} \cos\theta$
$\psi_{21\pm1}$	$\psi_{2p_x/2p_y} = C\left(\frac{Zr}{a_0}\right) e^{-Zr/2a_0} \sin\theta\, e^{\pm i\phi}$

C = Constant
a_0 = bohr
Z = atomic no.



$\left(2 - Zr/a_0\right) = 0$
$\Rightarrow r = 2a_0/Z$



Molecular Orbital Theory & Symmetry: Linear Combinations of Atomic Orbitals (LCAO)

$$\Psi_{\pm} = C_1\Psi_1 \pm C_2\Psi_2$$

C_1 & C_2 : Atomic Orbital Coefficients

Ψ_+ : $C_1\Psi_1 + C_2\Psi_2$ (bonding)

Ψ_- : $C_1\Psi_1 - C_2\Psi_2$ (anti-bonding)

Atomic Orbitals // H-like wavefunctions

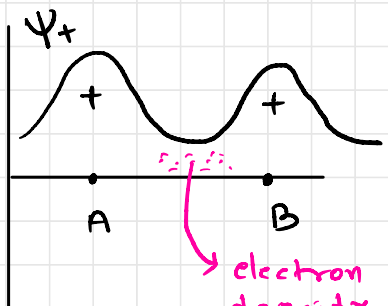
Molecular Orbitals

Two AOs (Ψ_1 & Ψ_2) generate two MOs (Ψ_+ & Ψ_-)

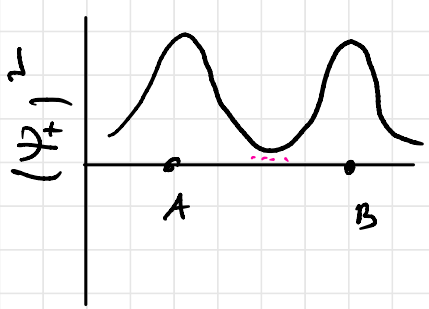
A Ψ_{1s}^A B Ψ_{1s}^B

Ψ_{1s} : Atomic Orbitals centered on Atoms

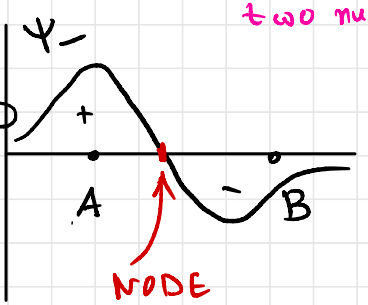
Bonding MO

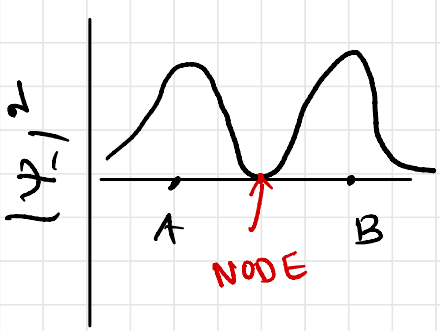


electron density between two nuclei

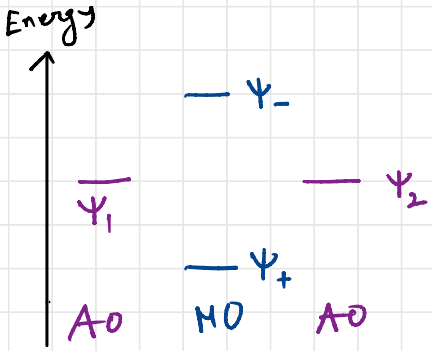


Antibonding MO



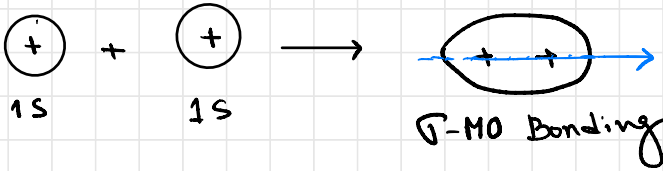


MO energy diagram

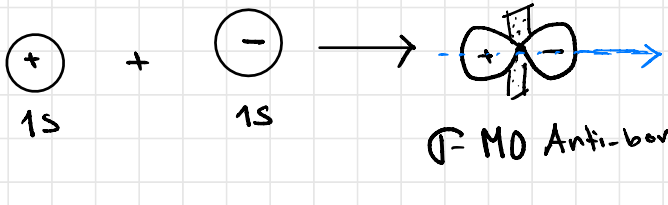


σ - MOs : Cyllendrically symmetric about the Molecular axis

A —— B → Molecular axis



g



u

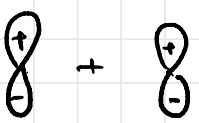
An MO is symmetric to the centre of inversion operation / symmetry →

Gerade
"g"
Even
Parity / symmetry

An MO is asymmetric to the centre of inversion operation / symmetry →

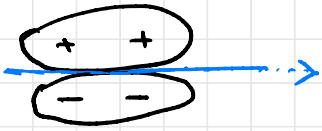
Ungerade
"u"
Odd
parity / symmetry

π - MOs : Nodal plane contains Molecular axis



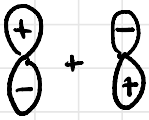
p_x p_x

AOs



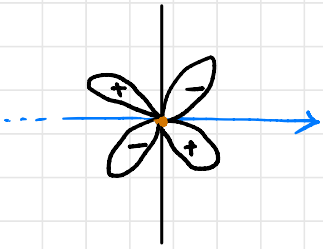
π - MO bonding

u



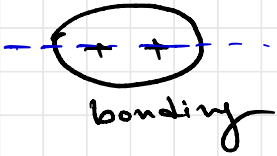
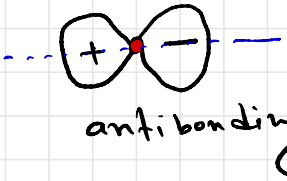
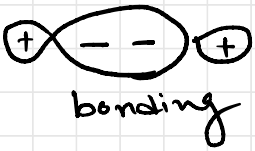
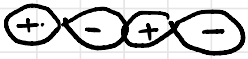
p_x p_x

AOs



π - MO Anti-bonding

g

AO 1	AO 2	MO	Orbital Symmetry
 1s	 1s	 bonding	σ_g
 1s	 1s	 antibonding	σ_u
 $2p_z$	 $2p_z$	 bonding	σ_g
 $2p_z$	 $2p_z$	 antibonding	σ_u